

# Acute Stabilization Pathway Diabetic Ketoacidosis: Outpatient Pathway

PEDIATRIC CARE



## Definition of DKA requires all of the following:<sup>1</sup>

|                      |                                                                                      |
|----------------------|--------------------------------------------------------------------------------------|
| <b>Hyperglycemia</b> | Glucose > 200 mg/dL                                                                  |
| <b>Ketosis</b>       | Blood $\beta$ -hydroxybutyrate > 3 mmol/L <b>or</b><br>"moderate or large" ketonuria |
| <b>Acidosis</b>      | Venous pH < 7.3 <b>or</b><br>Serum $\text{HCO}_3^-$ < 15 mmol/L                      |

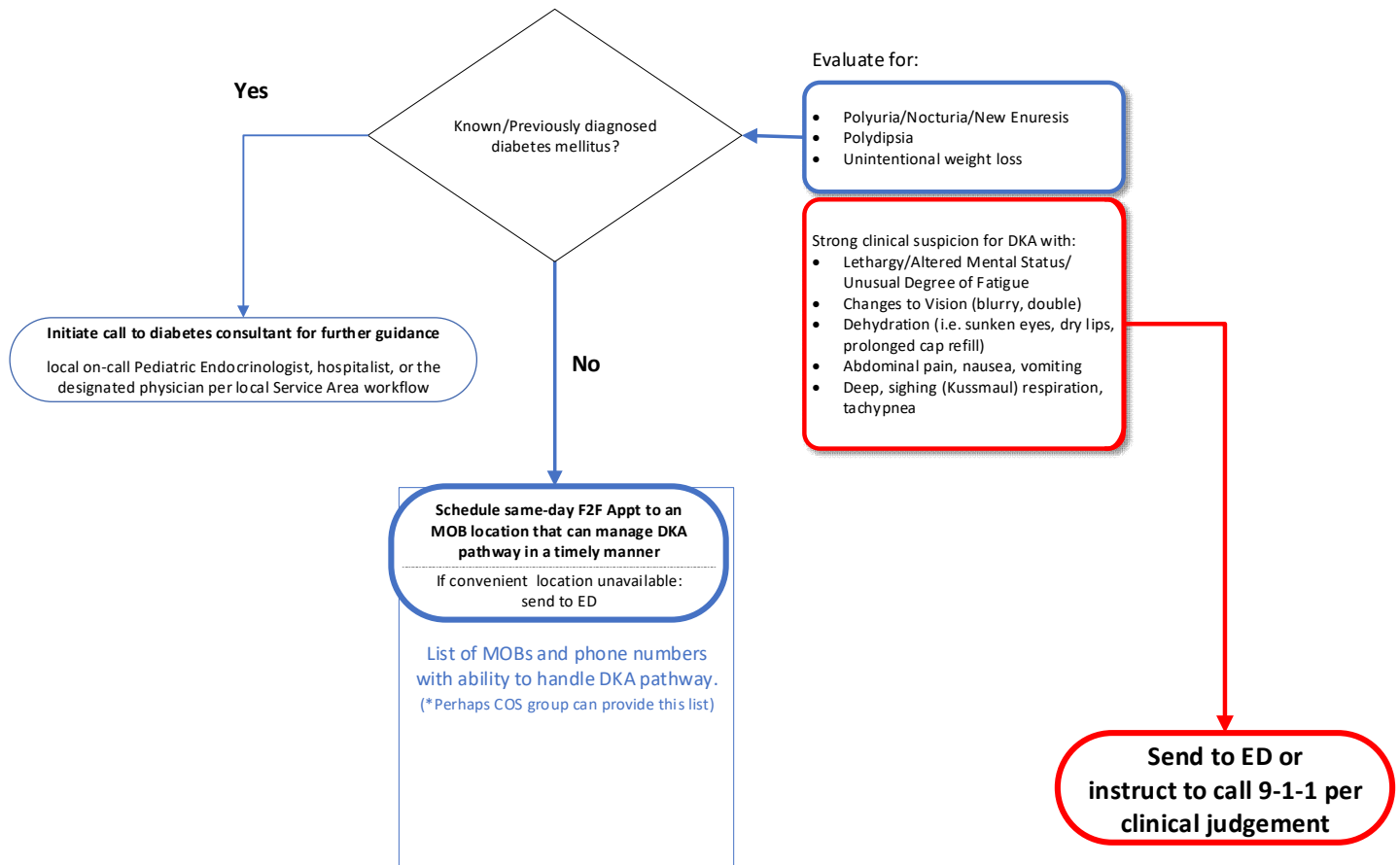
## Goals of Therapy<sup>2</sup>

|                                  |                                                                                                                                                                                                                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Correct Dehydration</b>       | Conservative fluid resuscitation starting with 10 mL/kg bolus of normal saline<br>Careful reassessment to avoid contributing to risk of cerebral edema                                                                                                                                          |
| <b>Correct Ketoacidosis</b>      | Start insulin AFTER the initial hour of fluid resuscitation <sup>3</sup><br>Do not use insulin bolus, it increases the risk for cerebral edema <sup>3</sup><br>$\text{HCO}_3^-$ > 15 mmol/L paired with anion gap closure alone may be used to determine DKA resolution. <sup>4</sup>           |
| <b>Normalize Blood Sugar</b>     | Aim to drop glucose by 50-100 mg/dL per hour                                                                                                                                                                                                                                                    |
| <b>Avoid Complications</b>       | Anticipate and evaluate for cerebral edema<br>Assess mental status and neurologic exam frequently during treatment                                                                                                                                                                              |
| <b>Treat Precipitating Event</b> | Evaluate for underlying cause (infection, trauma, ingestion)                                                                                                                                                                                                                                    |
| <b>Important Note</b>            | In the outpatient setting without a stat lab, it is generally impossible to fully make a diagnosis of DKA. Thus, it is necessary based on symptoms and simple tests (POCT BG and UA) to determine the proper disposition so these goals of treatment can be safely and effectively carried out. |



## Virtual Visit

\*Consider diabetes mellitus in the differential diagnosis of children with any of the listed signs/symptoms.

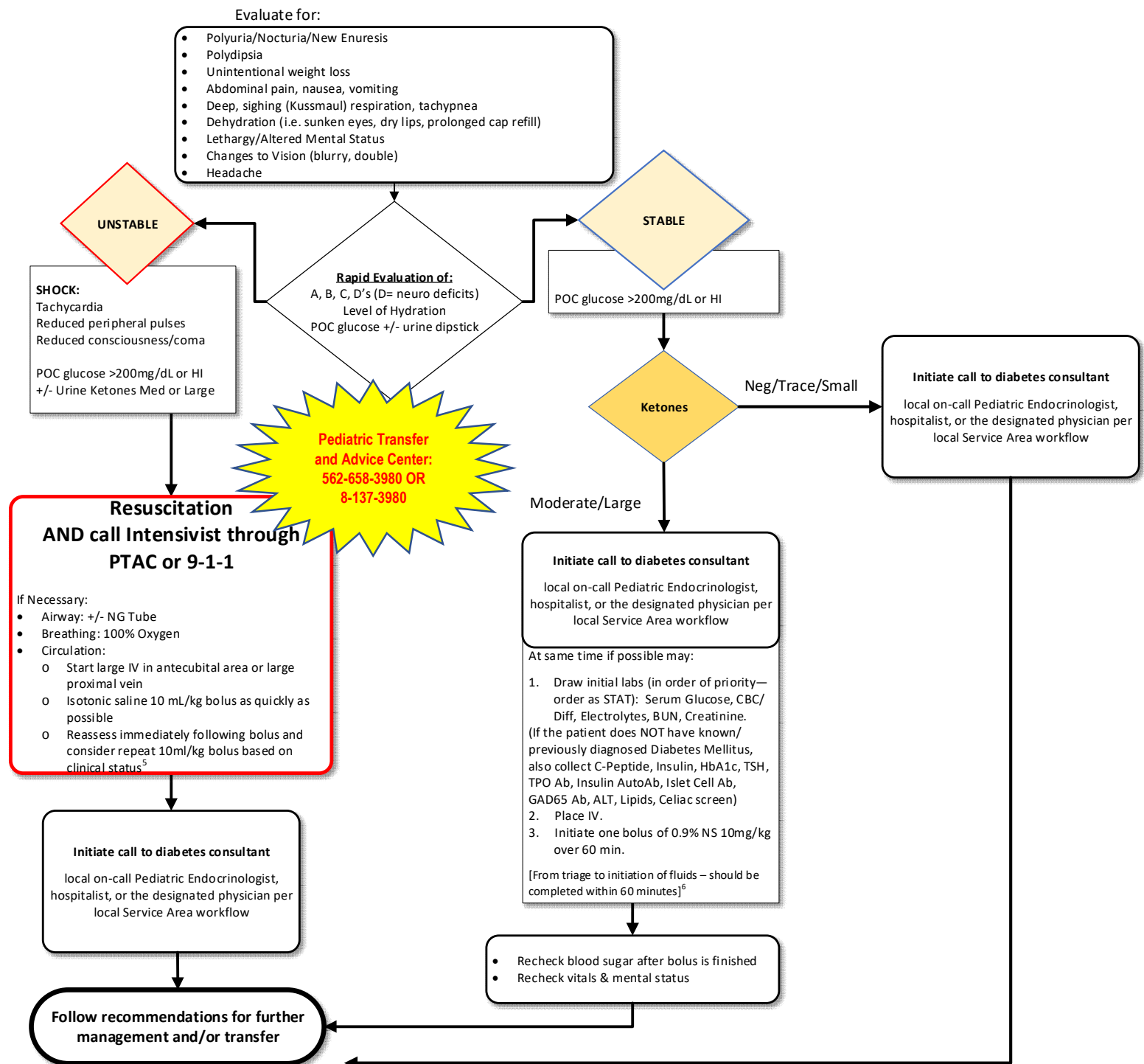


\* If there is concern for DKA, strongly advise against sending the patient to the lab without F2F appointment follow-up. If clinical judgement dictates sending to the lab, it is the responsibility of the ordering provider to be sure the patient goes to the next venue of care in a timely fashion and to follow up if they do not.



## Face To Face Visit

\*Consider diabetes mellitus in the differential diagnosis of children with any of the listed signs/symptoms.



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<sup>1</sup>Dunger DB, Sperling MA, Acerini CL, et al. ESPE/LWPES consensus statement on diabetic ketoacidosis in children and adolescents. *Arch Dis Child*. 2004;89(2):188-194.

<sup>2</sup>Wolfsdorf JI, Glaser N, Agus M, et al. ISPAD Clinical Practice Consensus Guidelines 2018: Diabetic ketoacidosis and the hyperglycemic hyperosmolar state. *Pediatric Diabetes*. October 2018; 19 (Suppl. 27): 155–177.

<sup>3</sup>Edge JA, Jakes RW, Roy Y, et al. The UK case-control study of cerebral oedema complicating diabetic ketoacidosis in children. *Diabetologia*. 2006;49(9):2002-2009.

<sup>4</sup> von Oettingen JE, Rhodes ET, Wolfsdorf JI. Resolution of ketoacidosis in children with new onset diabetes: Evaluation of various definitions. *Diabetes Res Clin Pract*. 2018;135:76-84.

<sup>5</sup> ISPAD Clinical Practice Consensus Guidelines 2018: Diabetic ketoacidosis and the hyperglycemic hyperosmolar state. Section 7.2.1.

<sup>6</sup> Accepted standard at multiple Children's Hospitals:

Children's Hospital of Philadelphia ED Pathway for the Evaluation/Treatment of the Child with Suspected DKA.

<https://www.chop.edu/clinical-pathway/diabetes-type1-with-dka-clinical-pathway>

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Seattle Children's Hospital Diabetic Ketoacidosis Pathway

<https://www.seattlechildrens.org/pdf/DKA-pathway.pdf>